

SHETH LUJ & MV COLLEGE

PRACTICAL NO 8

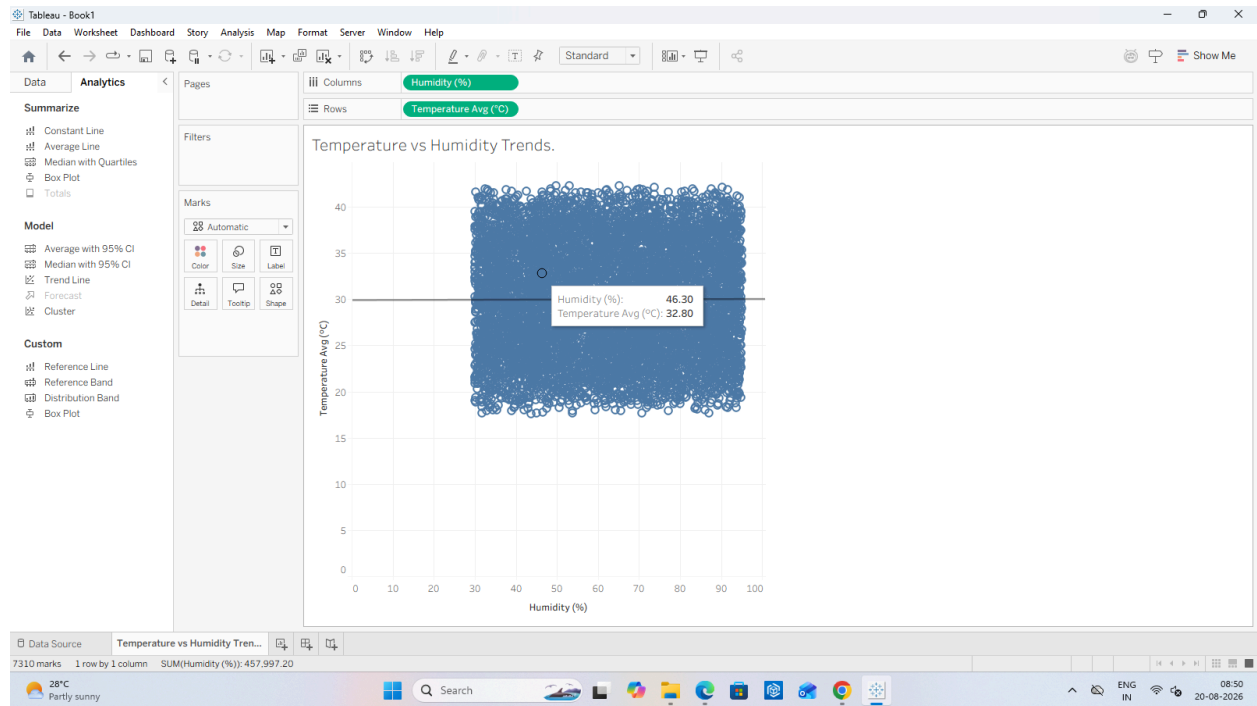
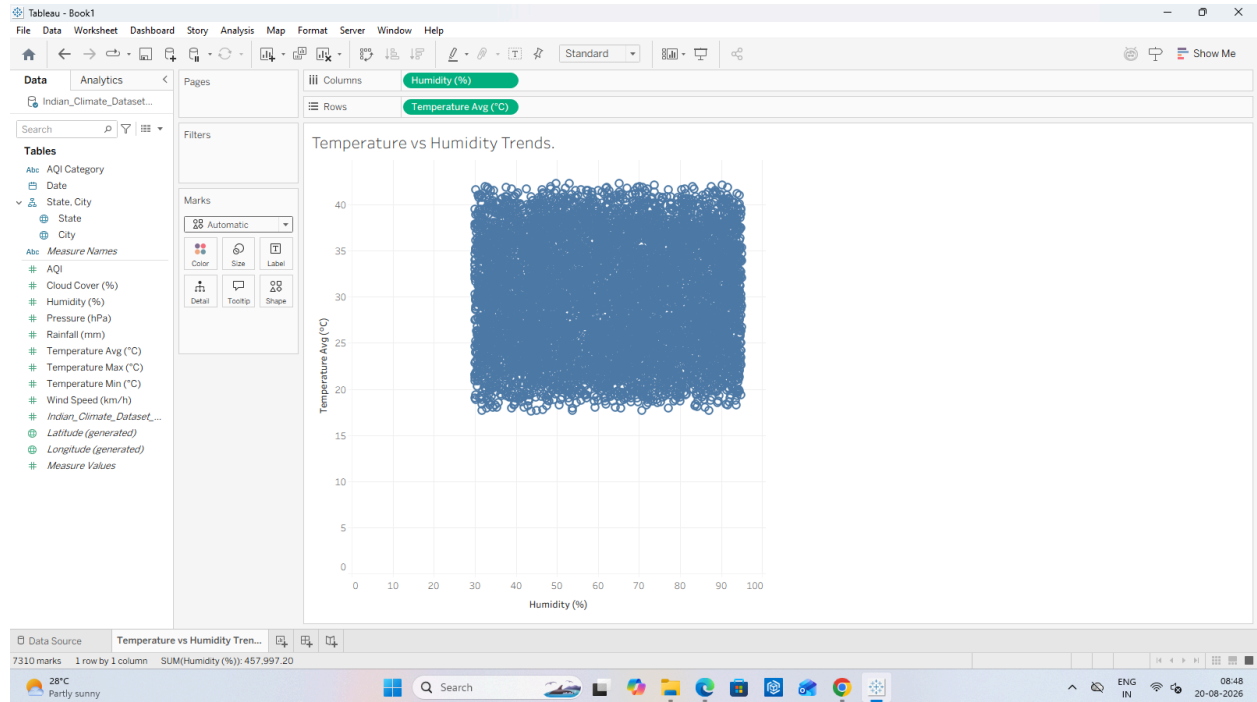
Aim: Performing Advanced Analytics using Trends, Clustering, and Forecasting

- Add and interpret trend lines (linear, exponential)
- Customize and analyze trend models
- Perform clustering on datasets
- Analyze distributions using built-in tools
- Apply forecasting techniques
- Compare different statistical models
- Export and interpret statistical results

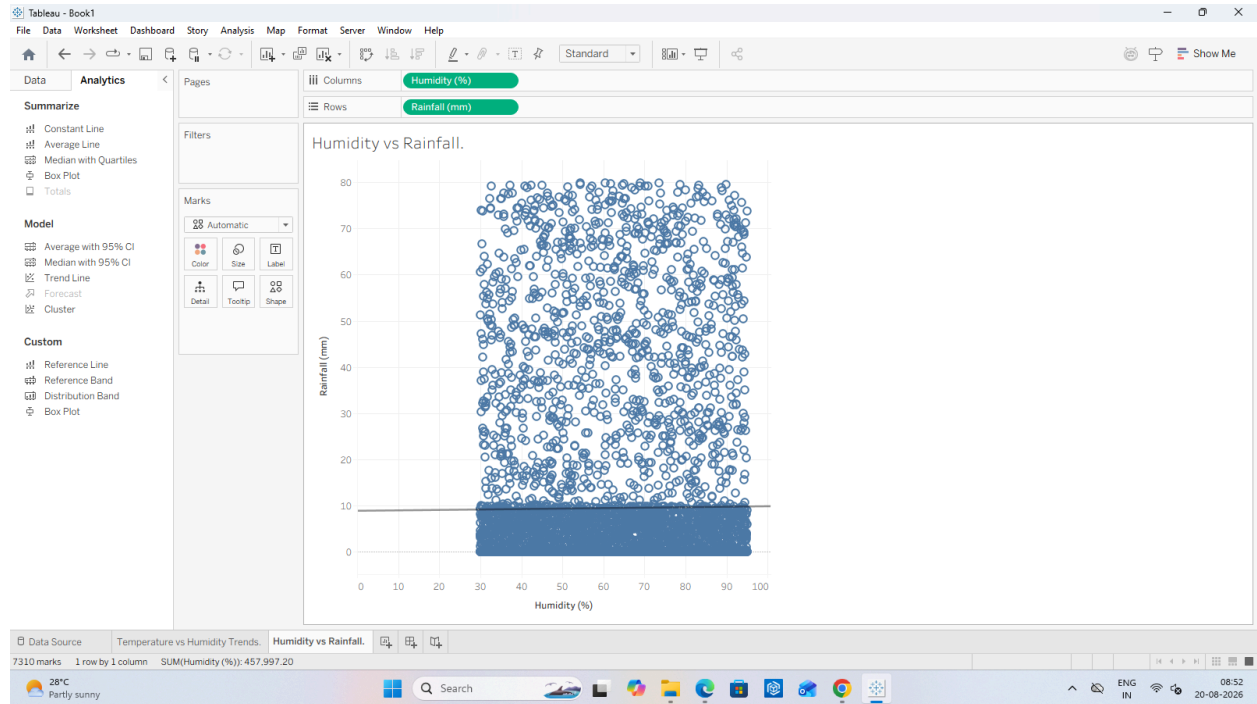
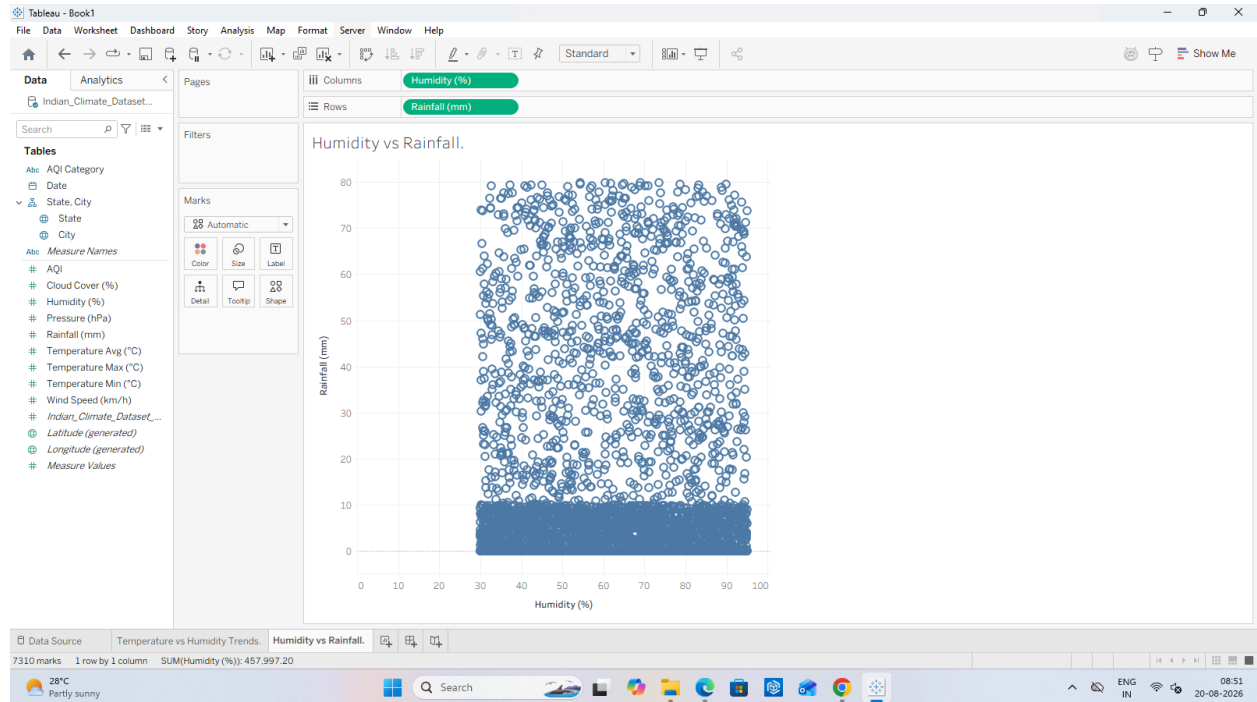
The screenshot displays the Tableau Desktop interface. The main view shows a dataset named 'Indian_Climate_Dataset_2024_2025' with 13 fields and 7310 rows. The fields are: Date, City, State, and Temperature Max (°C). The data is presented in a table format with columns for Date, City, State, and Temperature Max (°C). The interface includes a sidebar with 'Connections' and 'Files' sections, and a bottom status bar showing the system date and time.

Date	City	State	Temperature Max (°C)
01-01-2024	Mumbai	Maharashtra	32.5000
01-01-2024	Delhi	Delhi	25.4000
01-01-2024	Bengaluru	Karnataka	37.2000
01-01-2024	Chennai	Tamil Nadu	37.2000
01-01-2024	Kolkata	West Bengal	27.4000
01-01-2024	Hyderabad	Telangana	44.4000
01-01-2024	Ahmedabad	Gujarat	32.8000

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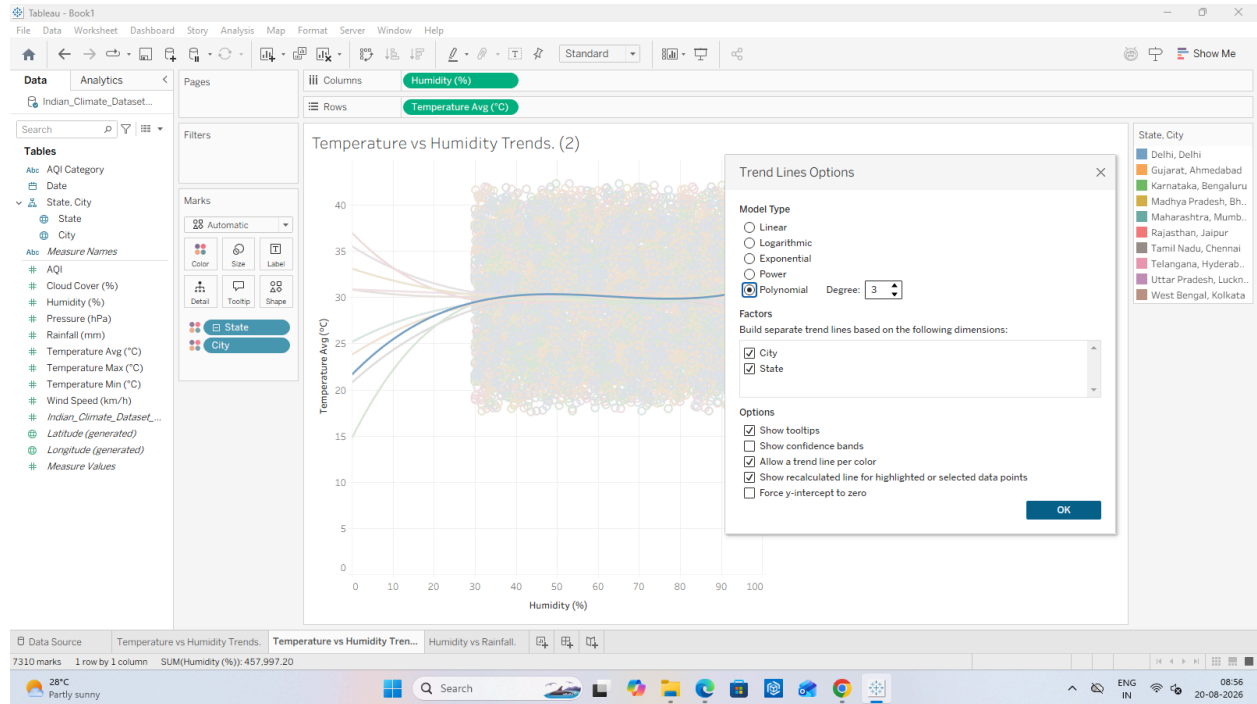
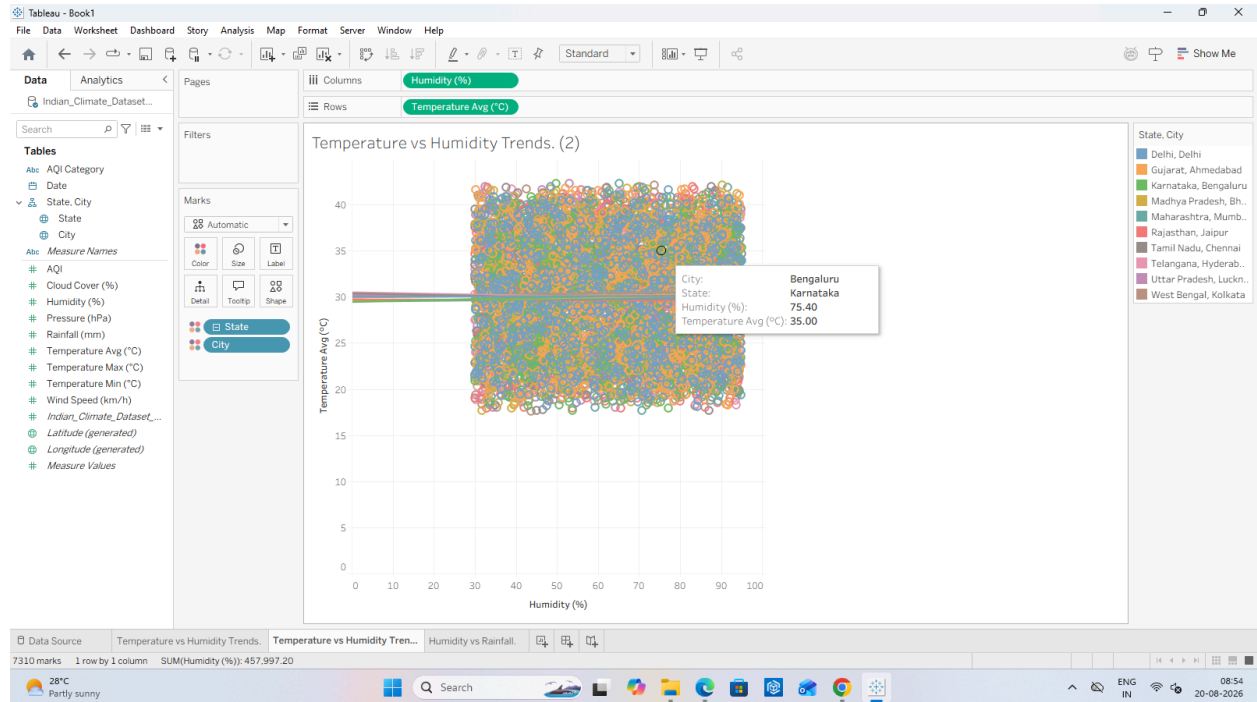


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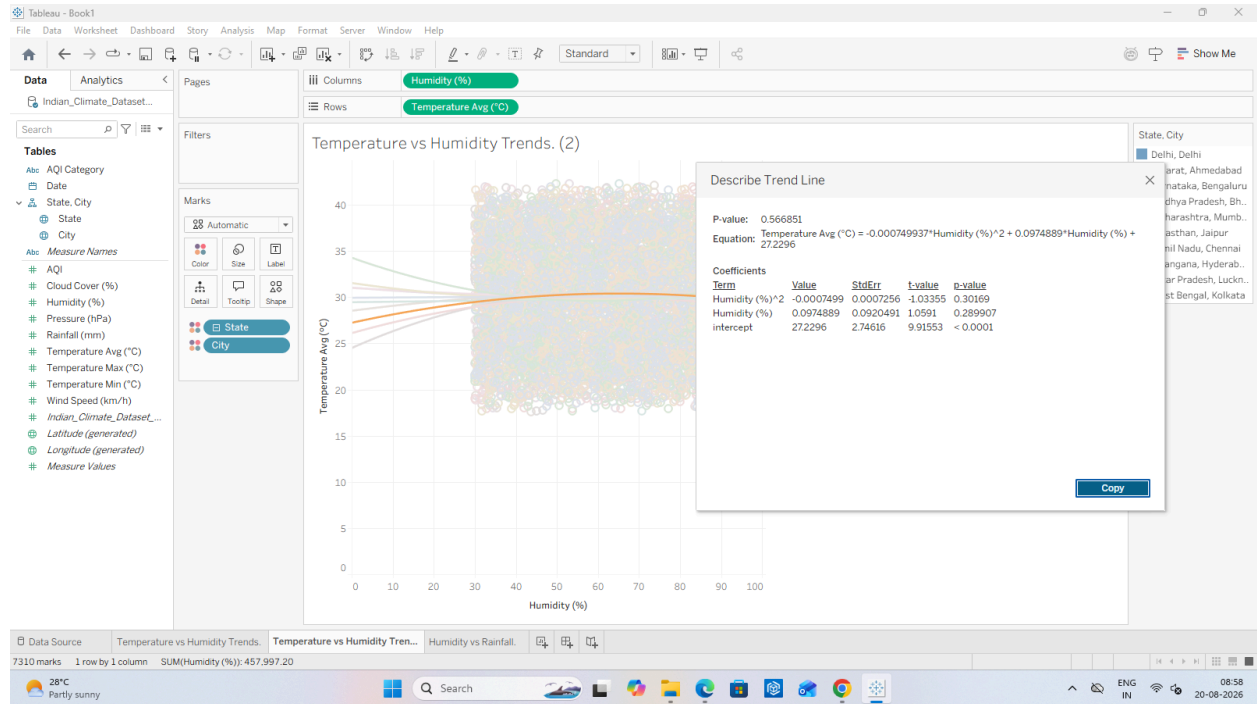
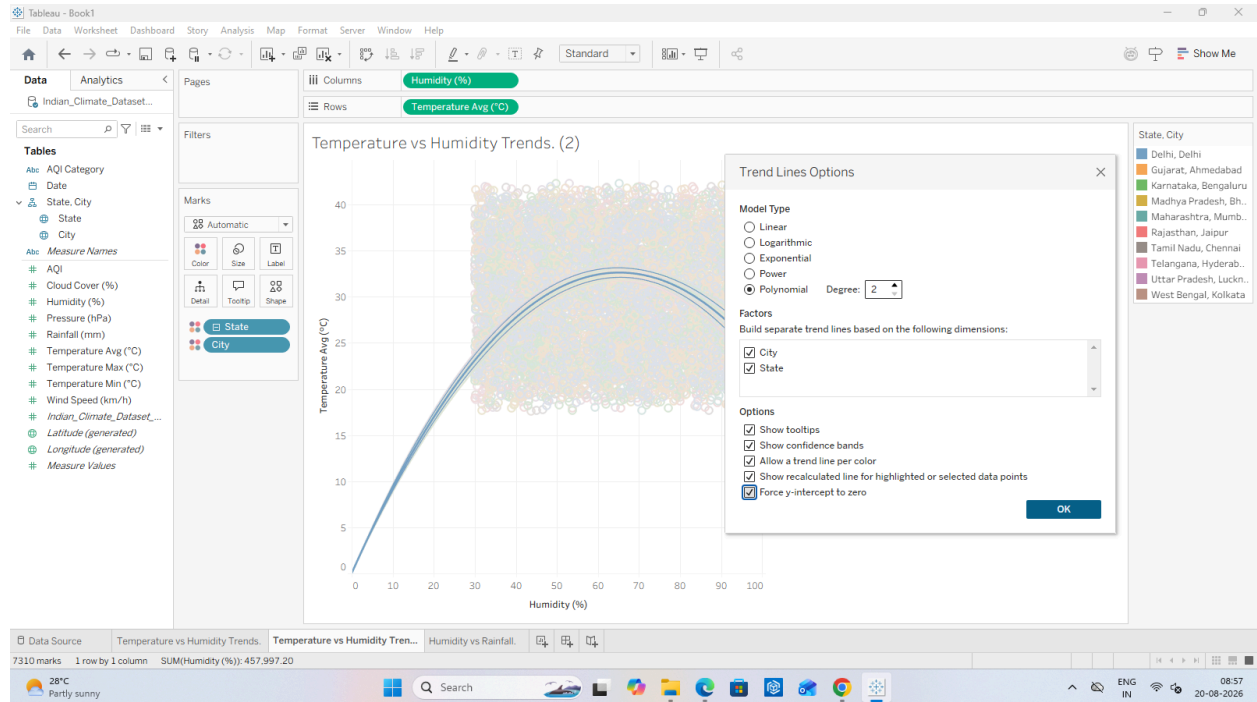


b) Customize and analyze trend models
OUTPUT:

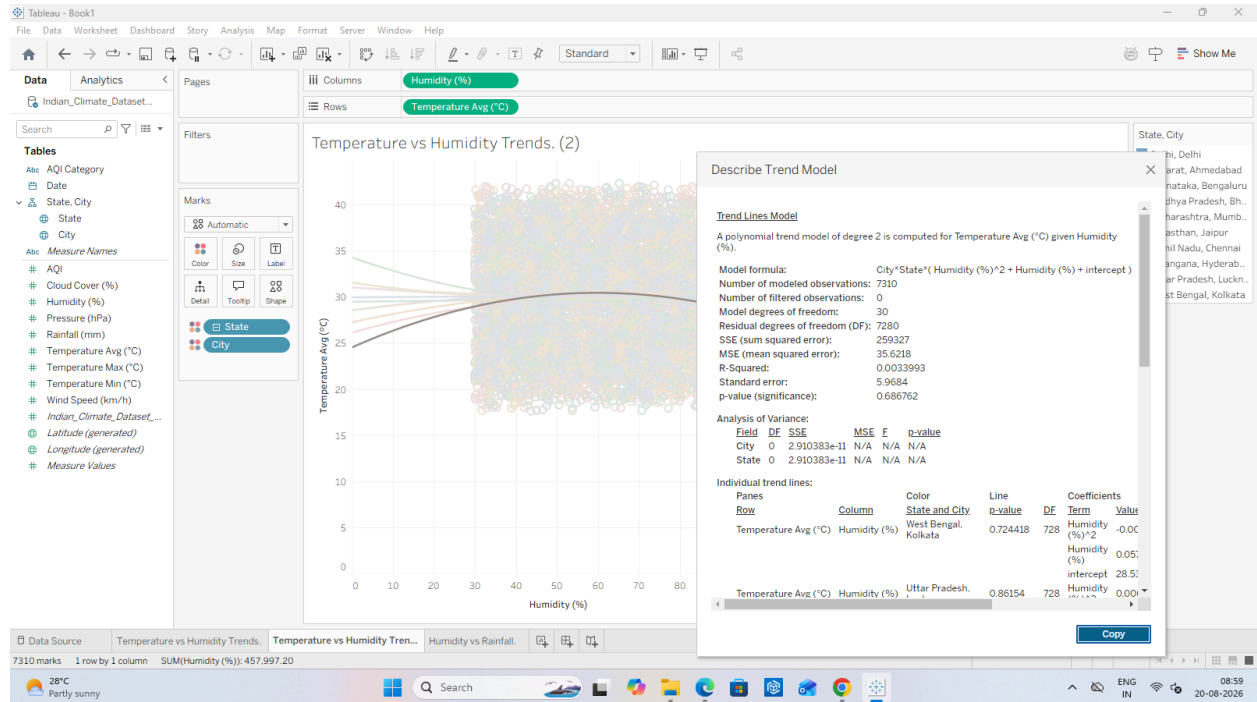
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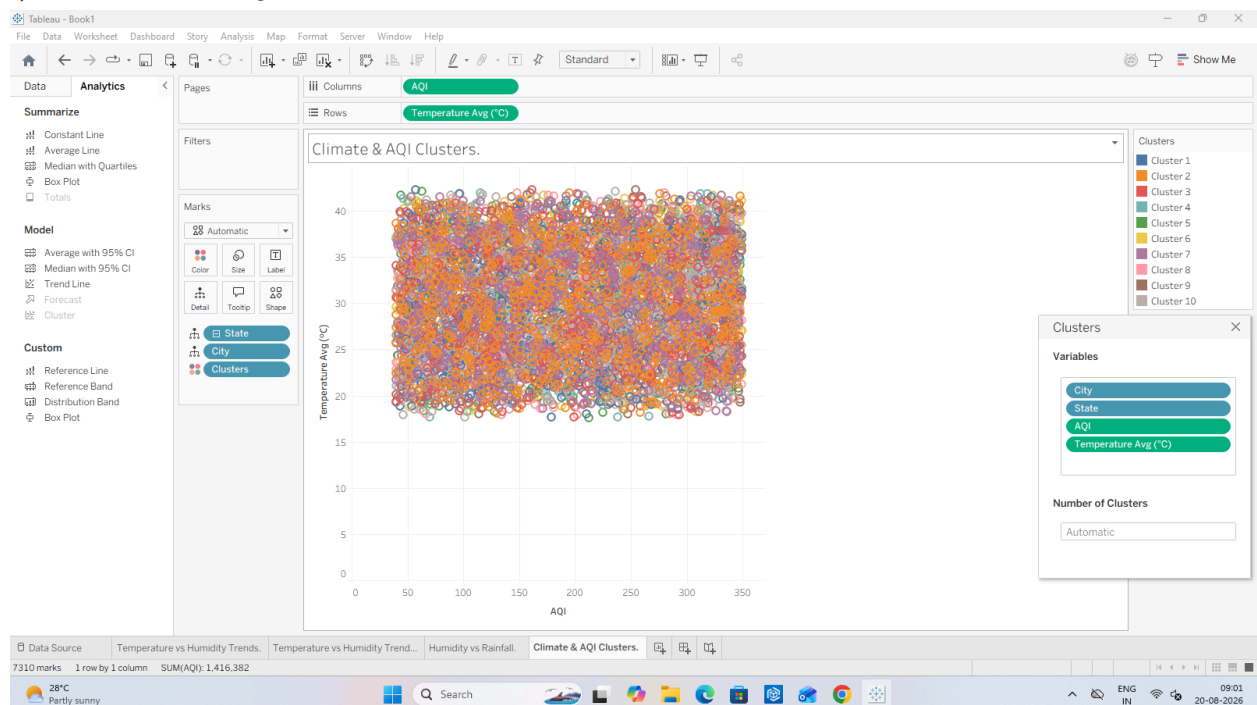
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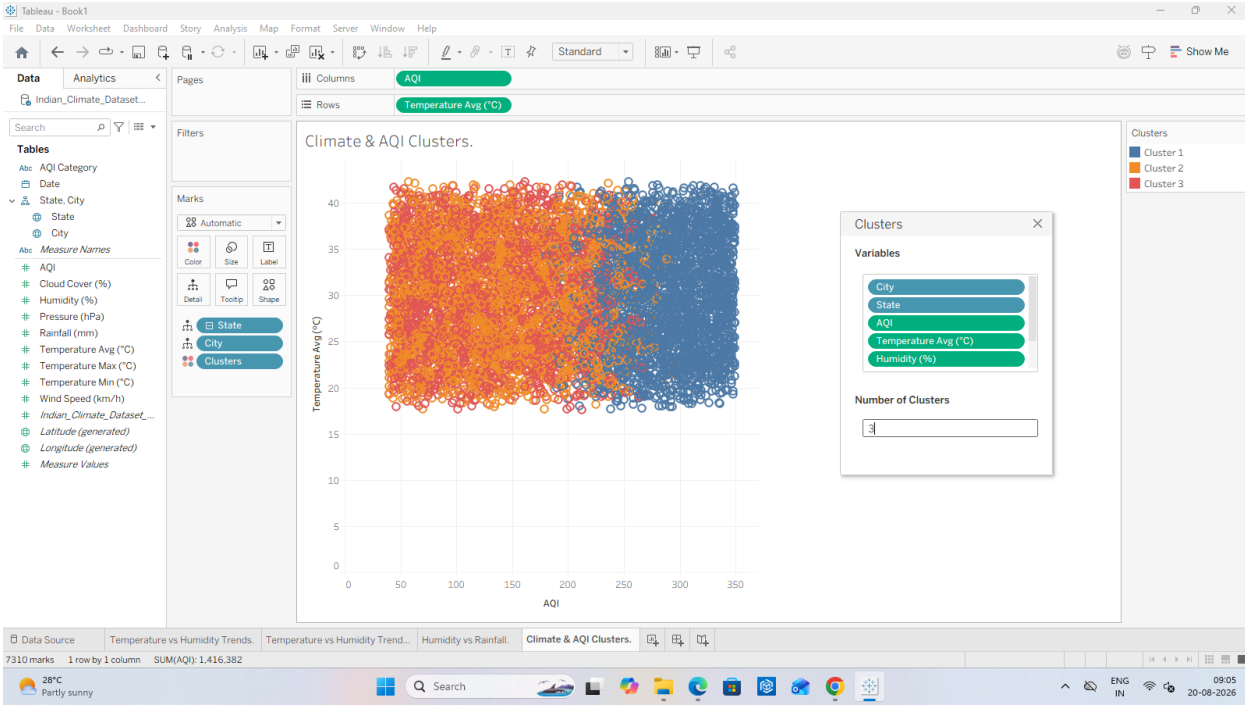
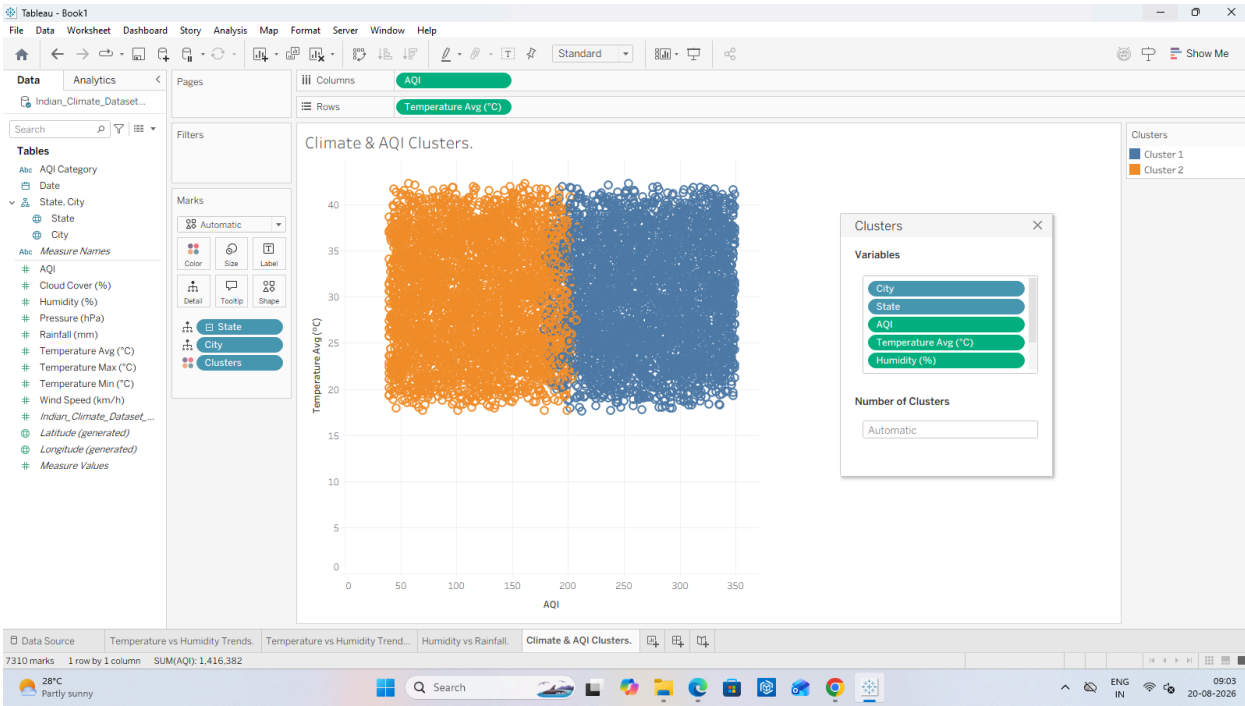
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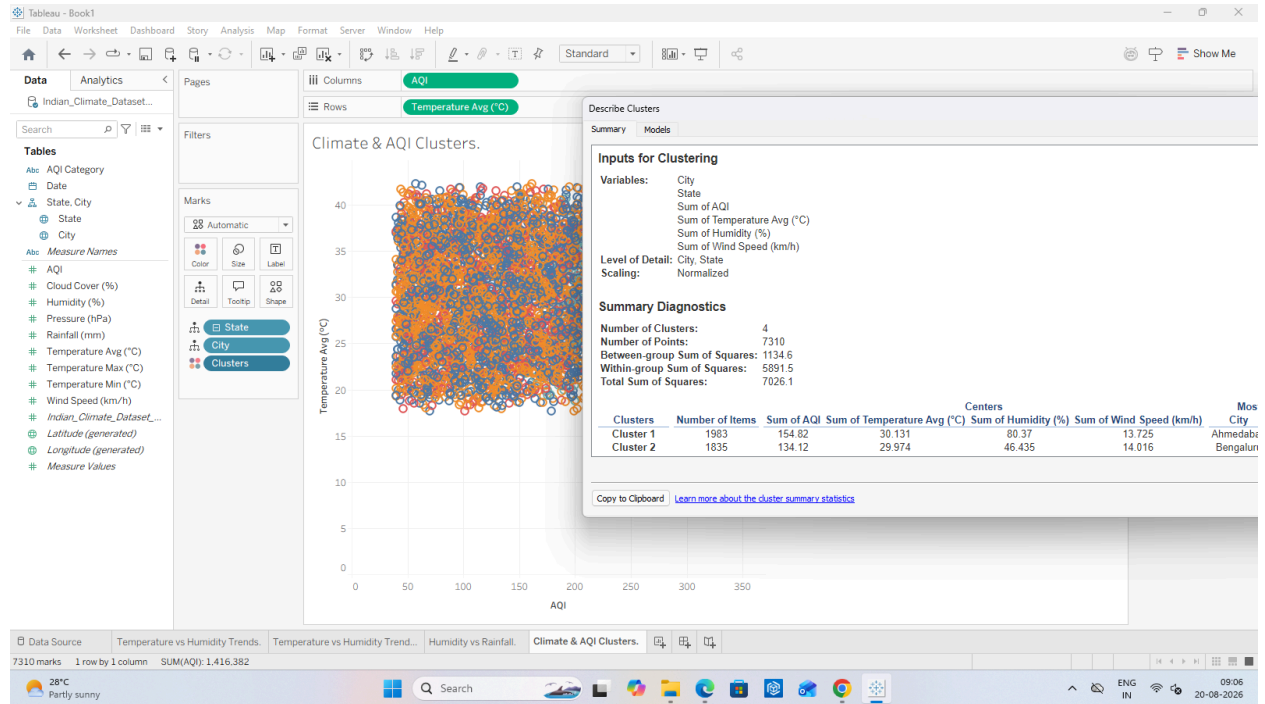
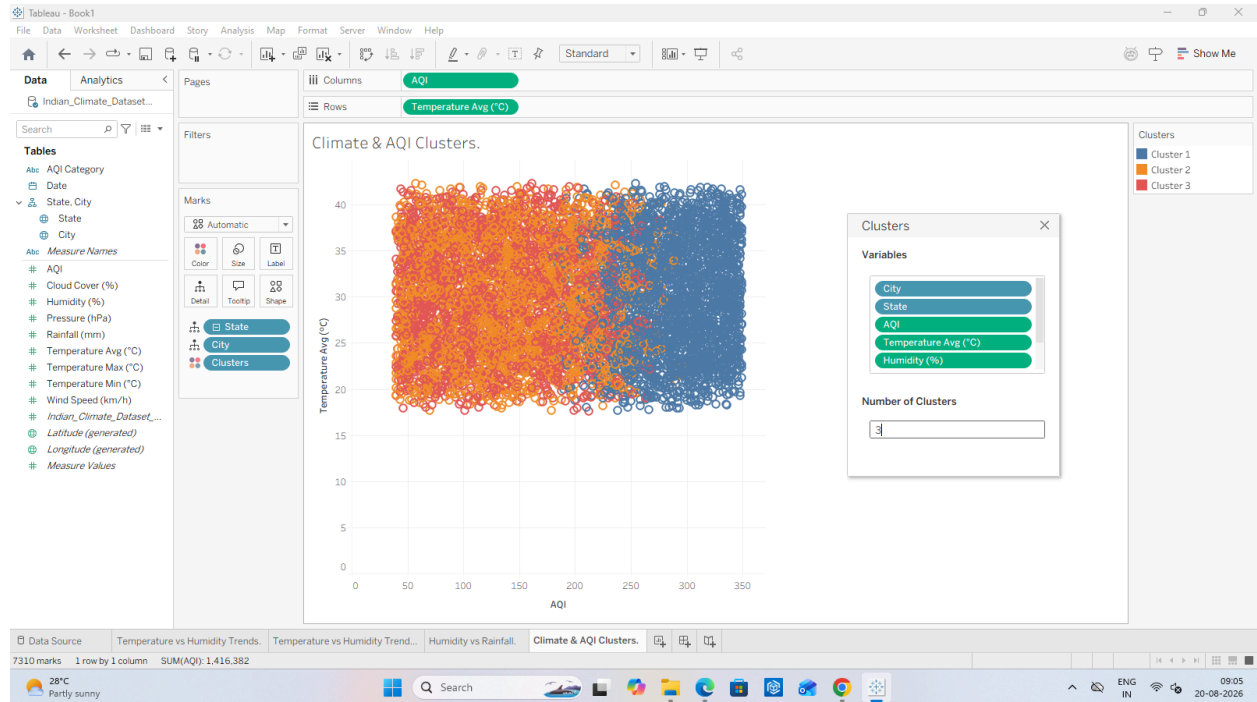
c) Perform clustering on datasets



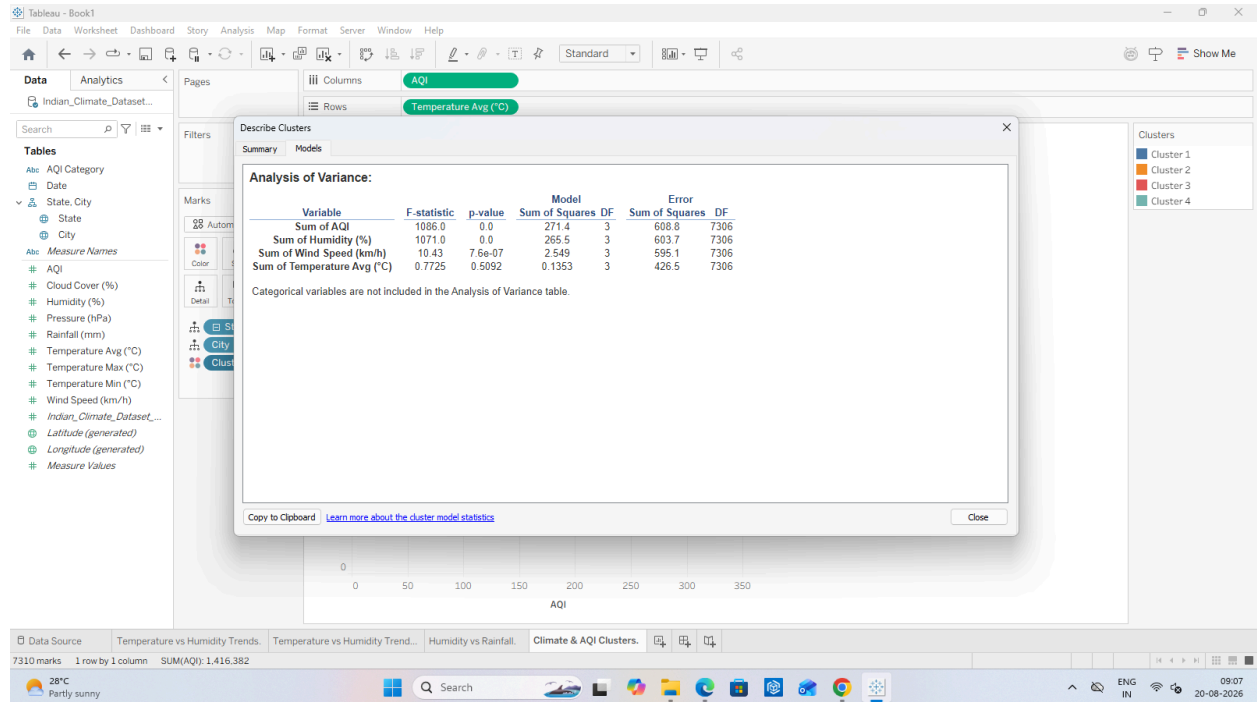
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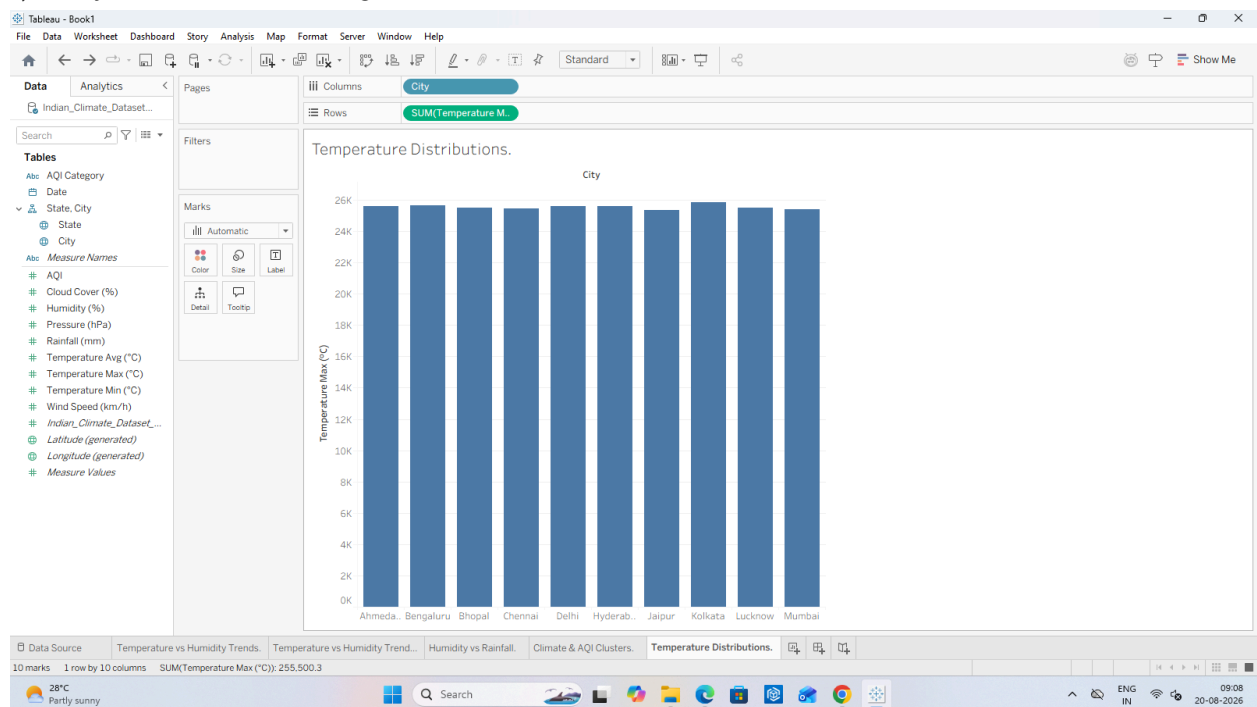
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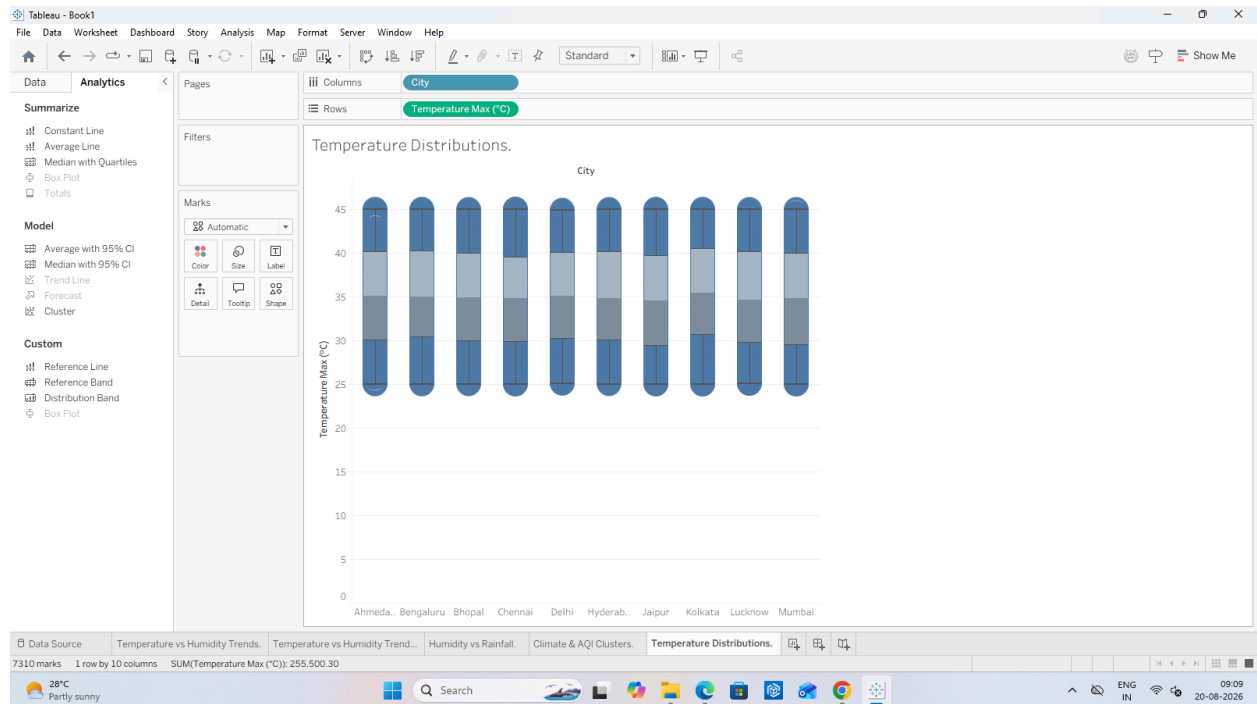
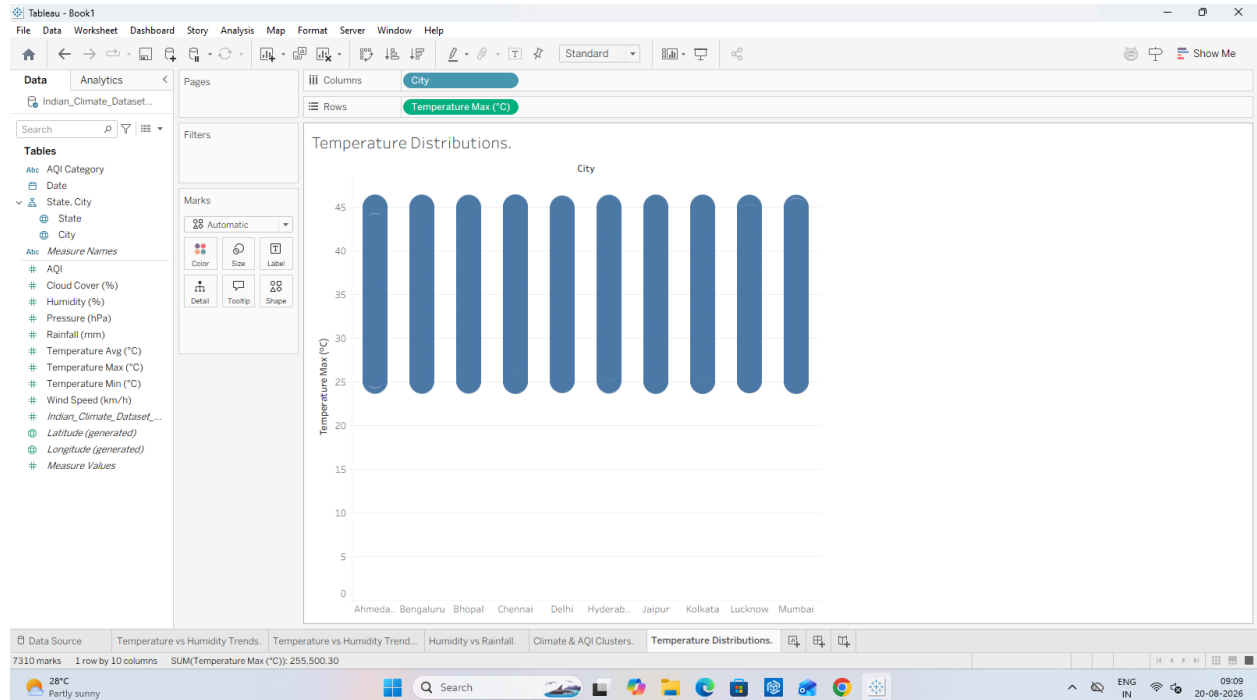
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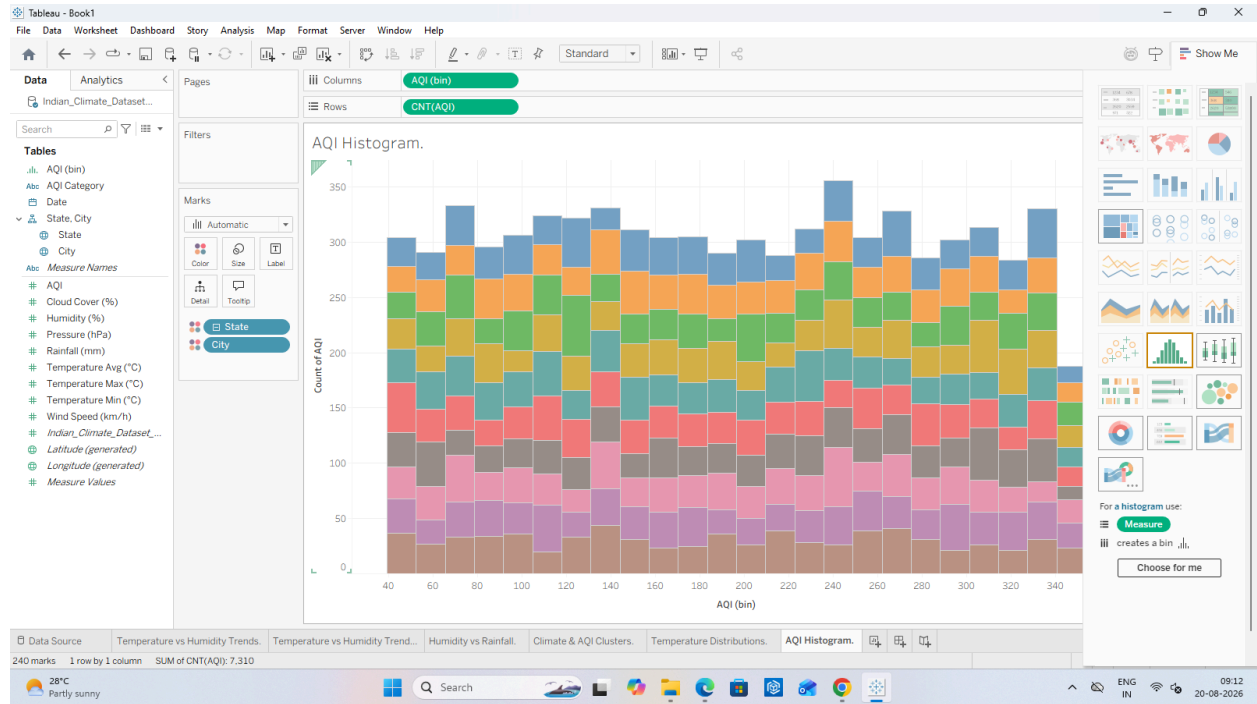
d) Analyze distributions using built-in tools



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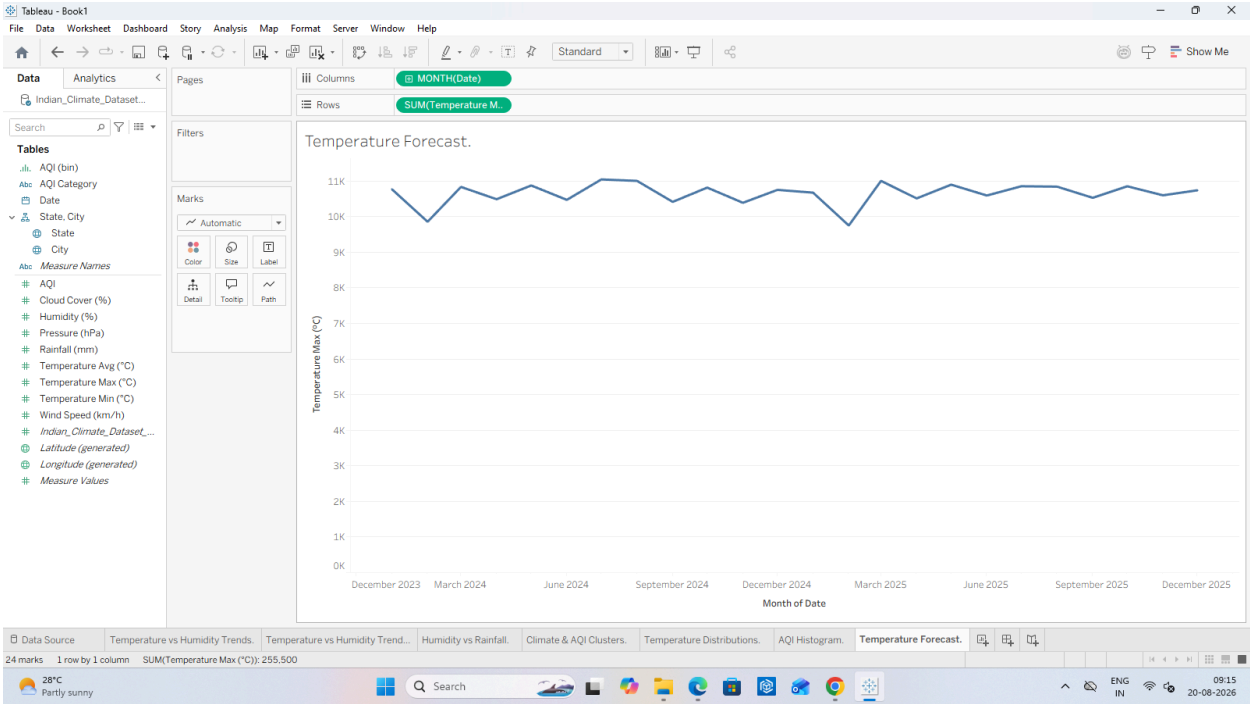
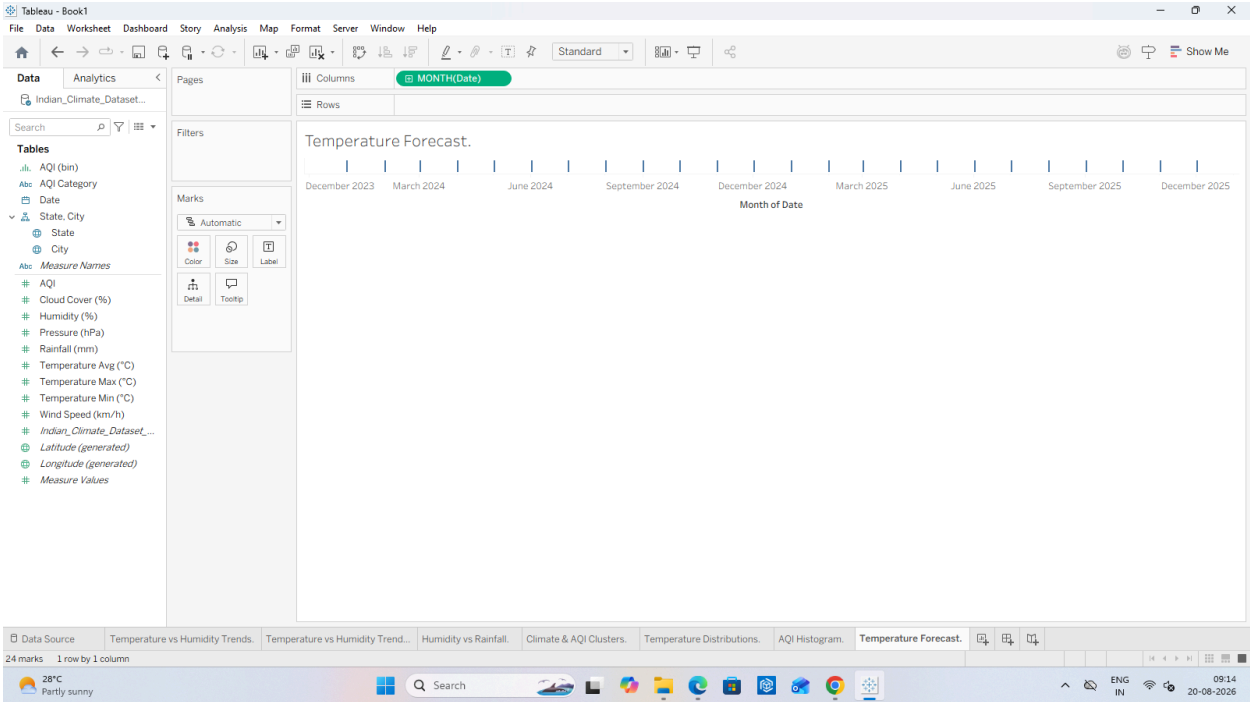


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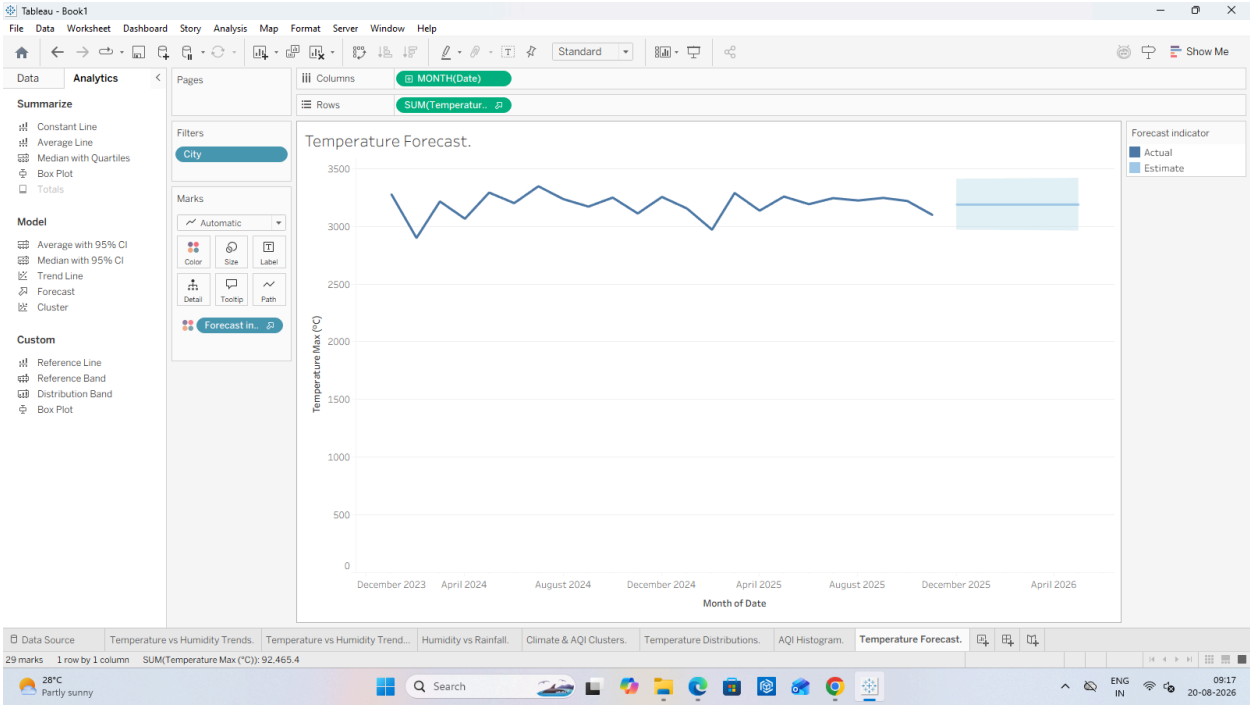
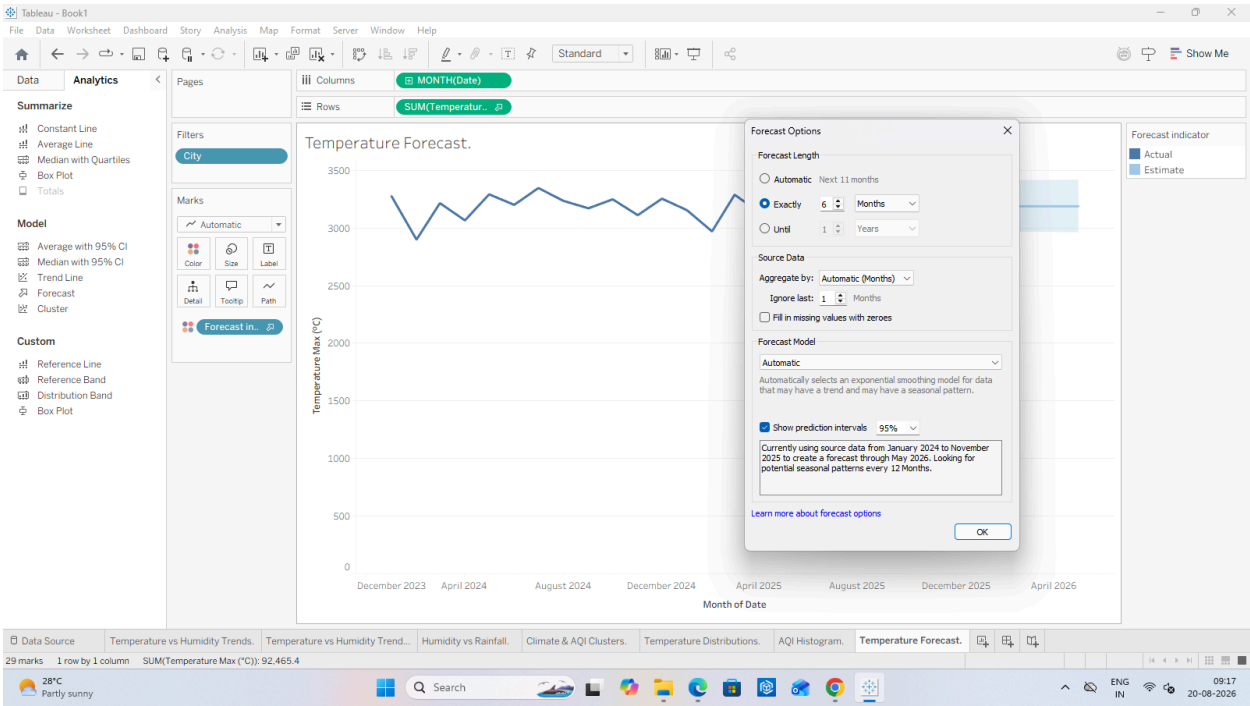


e) Apply forecasting techniques

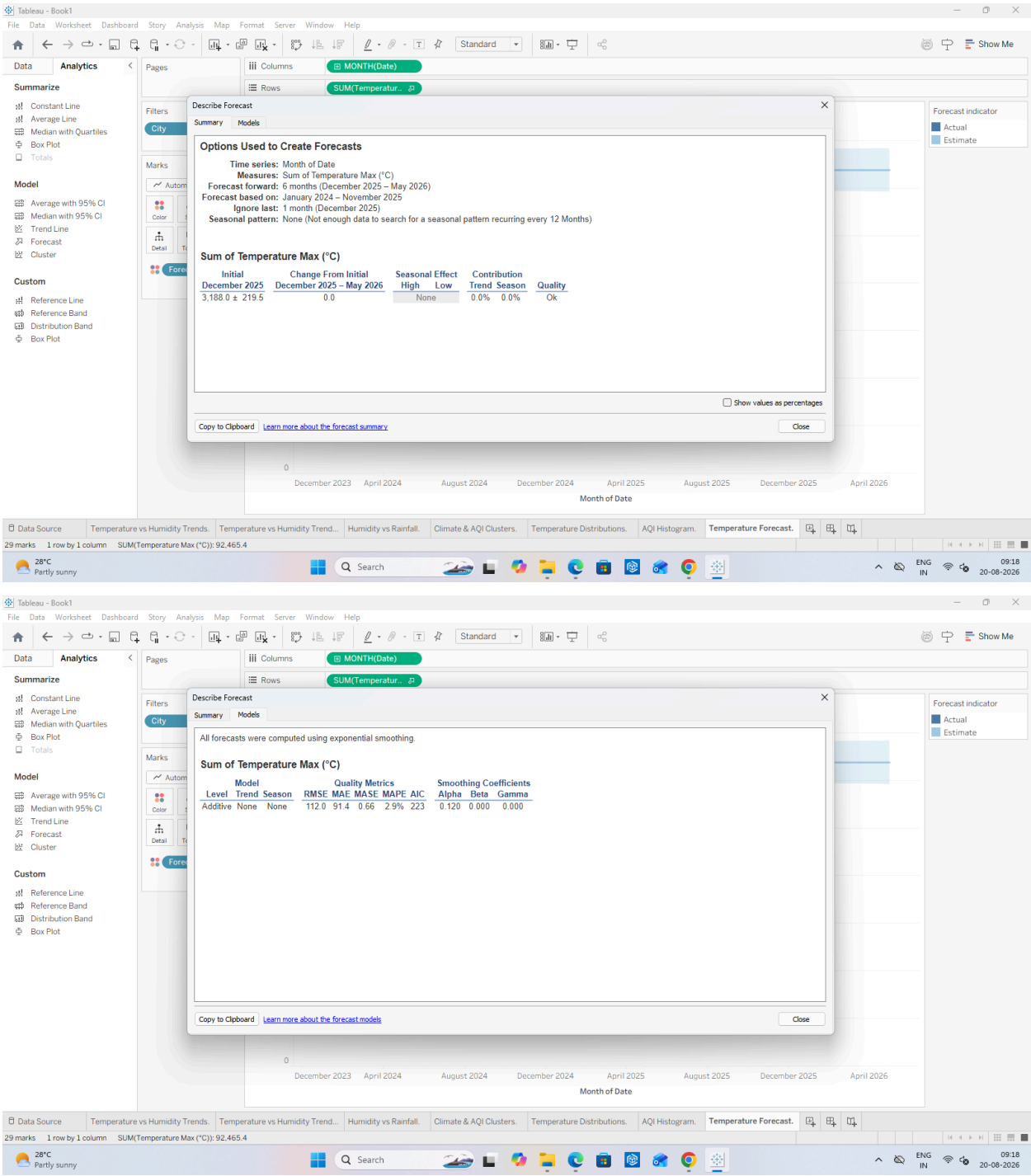
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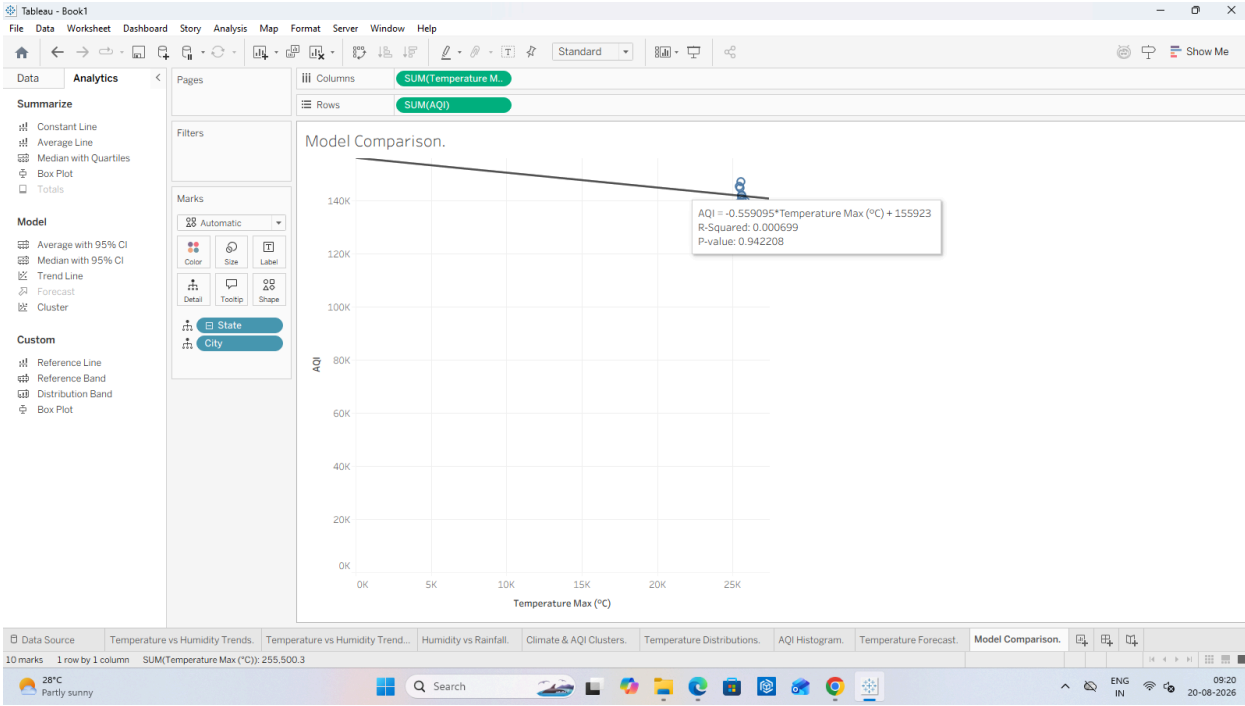
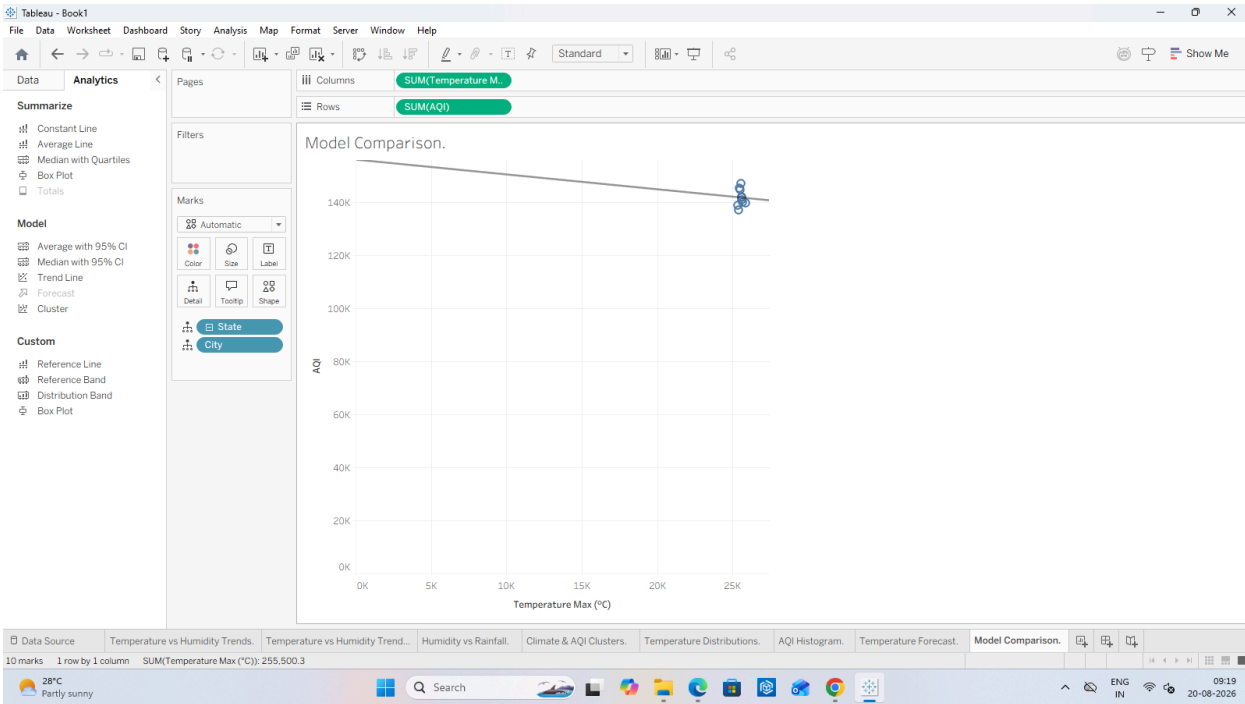
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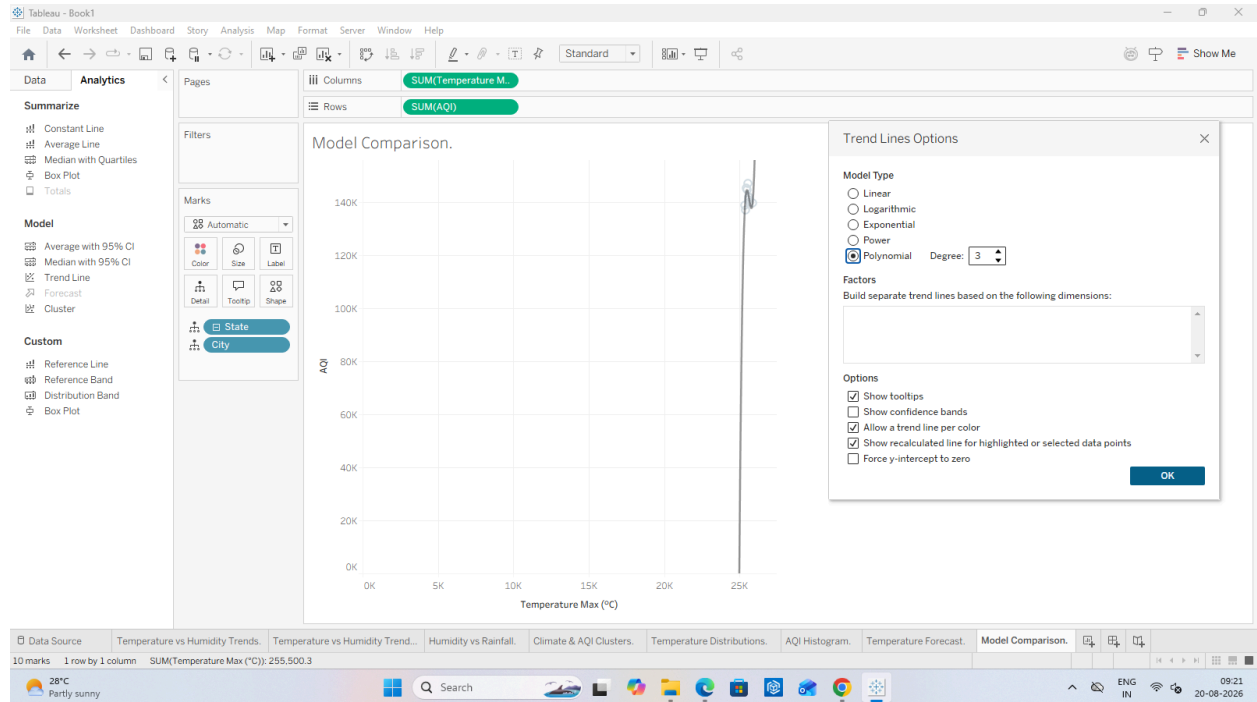
f) Compare different statistical models

PUSHPAL PATHAK T102

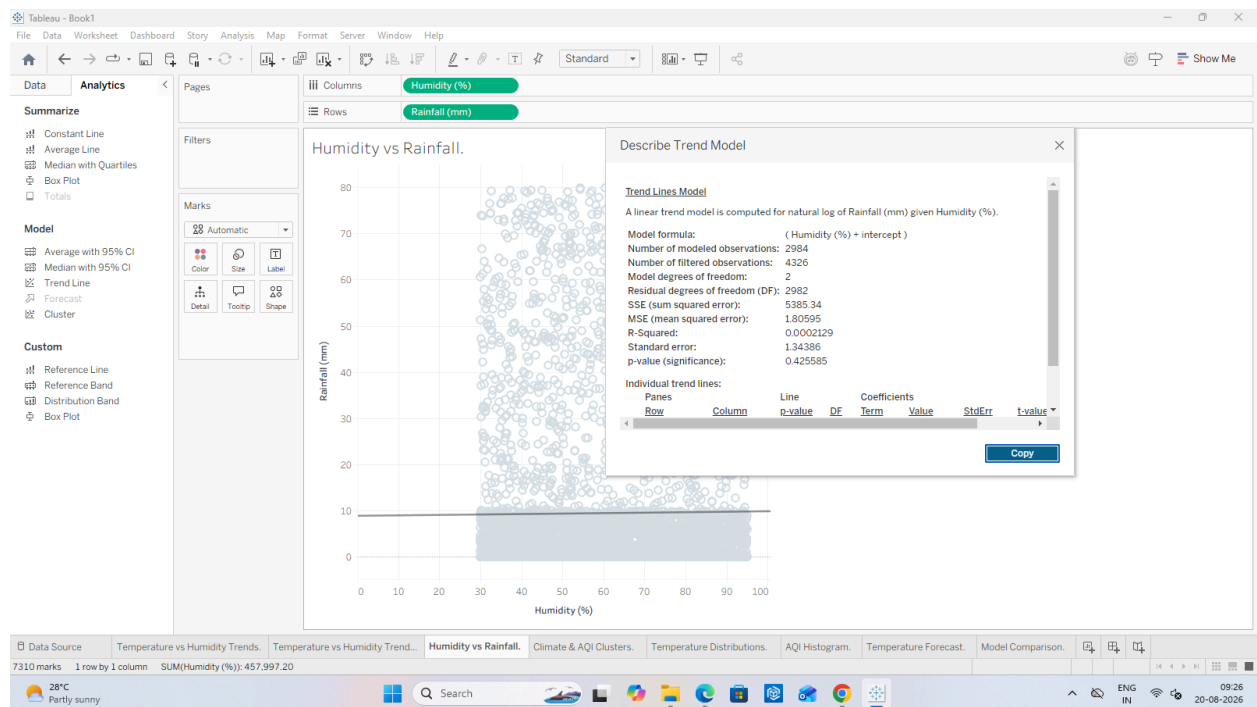
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g) Export and interpret statistical results



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```
Real estate.csv  diabetes_small.csv  INSTALL.txt  students.txt  Trend Lines Model  +
File Edit View H1 B I S G D
Trend Lines Model
A linear trend model is computed for natural log of Rainfall (mm) given Humidity (%).
Model formula: ( Humidity (%) + intercept )
Number of modeled observations: 2984
Number of filtered observations: 4326
Model degrees of freedom: 2
Residual degrees of freedom (DF): 2982
SSE (sum squared error): 5385.34
MSE (mean squared error): 1.80595
R-Squared: 0.0002129
Standard error: 1.34386
p-value (significance): 0.425585
Individual trend lines:
Panels Line Coefficients
Row Column p-value DF Term Value StdErr t-value p-value
Rainfall (mm) Humidity (%) 0.425585 2982 Humidity (%) 0.0010491 0.0013166 0.796879 0.425585
Intercept 2.18398 0.0854115 25.5701 < 0.0001
```

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28°C Partly sunny Search 09:25 20-08-2026